

Homework 06 Correlation

Due by 11:59pm, Friday, October 24, 11:59pm

S&DS 2300/5300/ENV 757

Submission Details

To make grading easier, please leave the following line of code in your assignment.

```
set.seed(1234)
```

1) Correlation of 2017 and 2018 New Haven Road Race Times.

On homework 5 you created a dataset called `nhcombined` that had times for runners who ran in both the 2017 and 2018 New Haven Road Race AFTER removing unusual observations. I've put the final resulting dataset HERE (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/nhcombined.csv).

(1.1). Read the data into an R object called `nhcombined`. Get the dimension and head of this data to make sure it has the same number of rows that you found in homework 5.

```
nhcombined <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/nhcombined.csv")
dim(nhcombined)
```

```
## [1] 982    6
```

```
head(nhcombined)
```

##	X	Name	Gender	Nettime_2018	Nettime_2017	improvement
## 1	1	Abbey Shaw	F	39.25000	40.25000	1.000000
## 2	2	Abby Dziura	F	39.03333	35.63333	-3.400000
## 3	3	Abby Ganun	F	40.08333	44.65000	4.566667
## 4	4	Abi Hawkins	F	35.86667	27.56667	-8.300000
## 5	5	Abigail Murphy	F	32.88333	34.06667	1.183333
## 6	6	Abraham Cordero	M	29.63333	31.83333	2.200000

(1.2) Find the code that defines the functions `myCor()` (class 11) and `permCor()` (class 12). Paste the code for both functions in a chunk below. Run each function on net times in 2018 and 2017 for runners in your `nhcombined` dataset to make a scatterplot of these times and to run a permutation test on the correlation between times (i.e. we're seeing if times from one year to the next are correlated within runners). Be sure to include the resulting histogram and the p-value of the test. Does it seem appropriate to calculate correlation (think about outliers)?

```

myCor <- function(x, y){
  #make the scatterplot
  plot(x, y, pch = 19, col = "red")
  #add text in the margin reporting the sample correlation
  mtext(paste("Sample Correlation =", round(cor(x, y), 3)), side = 3, cex = 1.2)
}

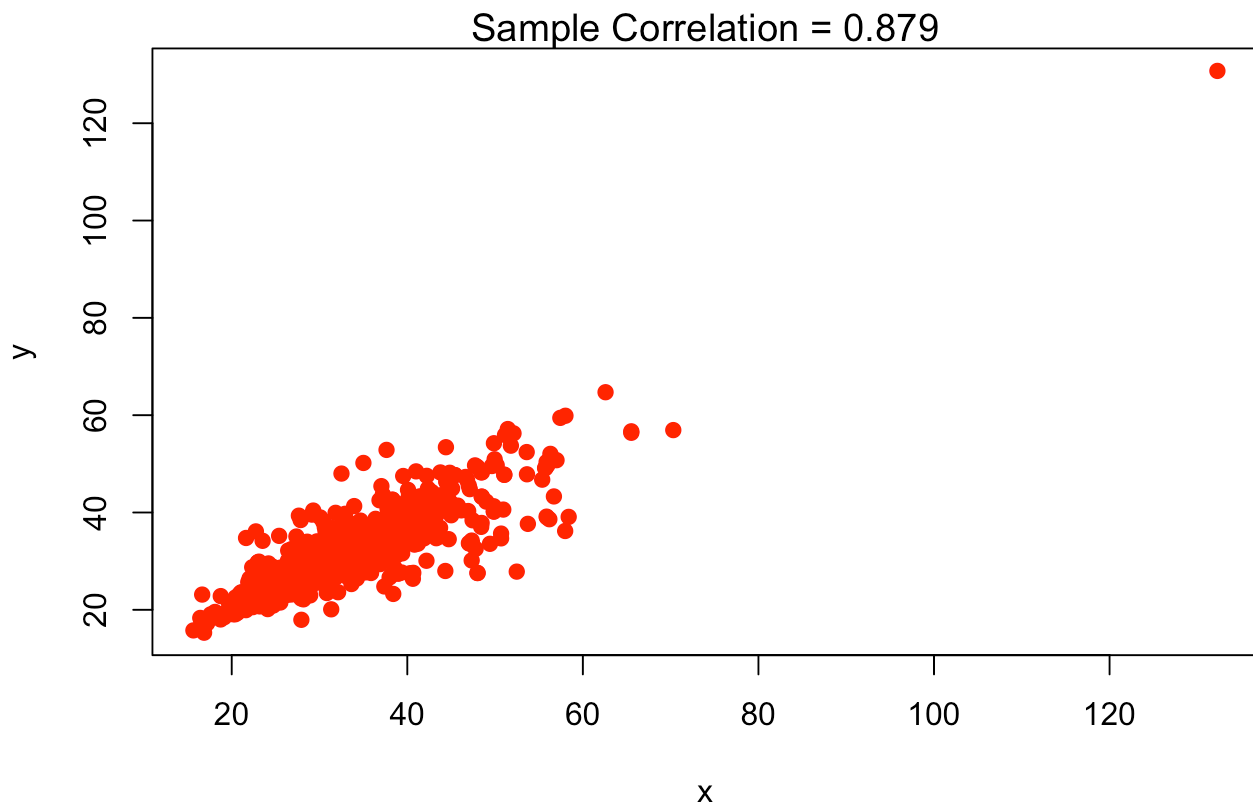
permCor <- function(x, y, n_samp = 10000, plotit = T){
  corResults <- rep(NA, n_samp)
  for (i in 1:n_samp){
    corResults[i] <- cor(x, sample(y))
  }

  pval <- mean(abs(corResults) >= abs(cor(x, y)))

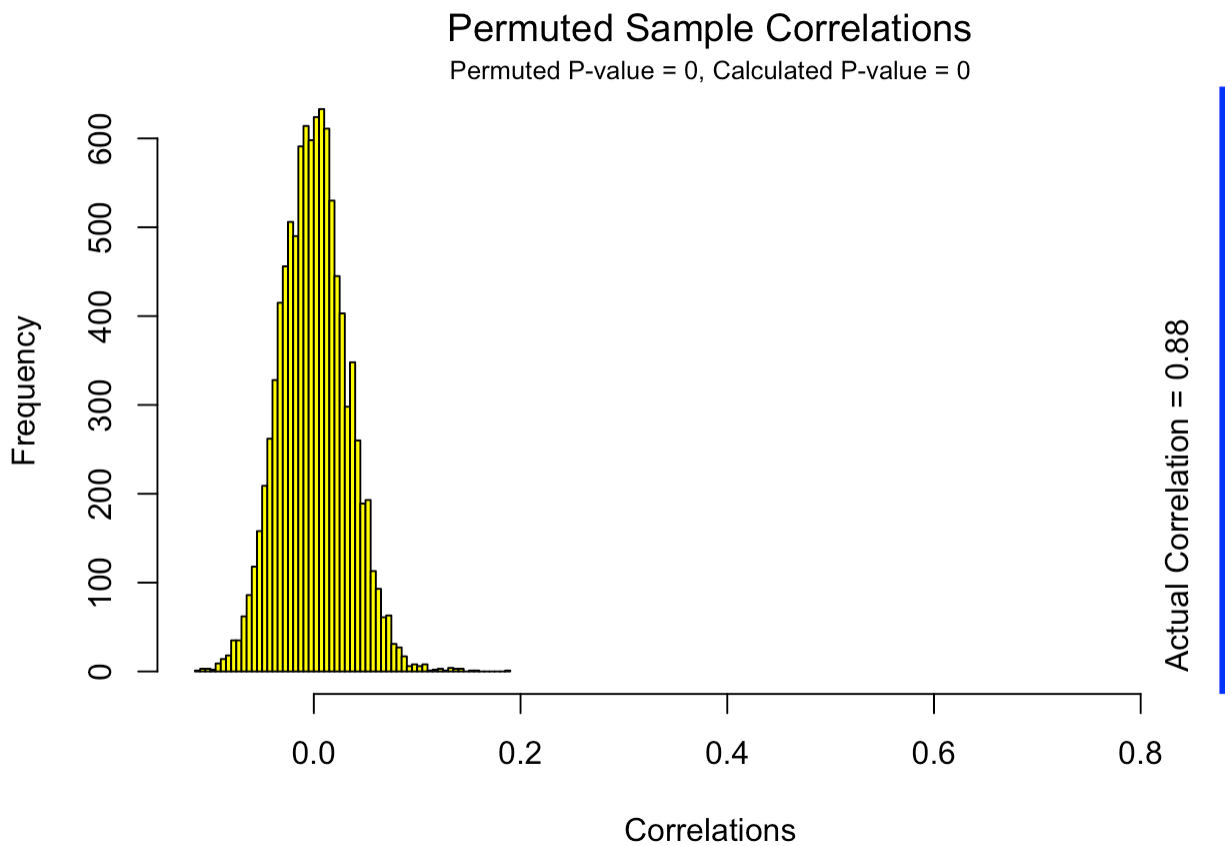
  if (plotit == T){
    #Make histogram of permuted correlations
    hist(corResults, col = "yellow", main = "", xlab = "Correlations", breaks = 50,
         xlim = range(corResults, cor(x, y)))
    mtext("Permuted Sample Correlations", cex = 1.2, line = 1)
    mtext(paste0("Permuted P-value = ", round(pval, 4), ", Calculated P-value = ", round(cor.test(x, y)$p.value, 4)), cex = 0.8, line = 0)
    abline(v = cor(x, y), col = "blue", lwd = 3)
    # Changed the location of the label for abline (from *1.1 to *0.95) to fit nicely onto the graph
    text(cor(x, y)*0.95, 0, paste("Actual Correlation =", round(cor(x, y), 2)), srt = 90, adj = 0)
  }
  if (plotit == F){
    return(round(pval, 5))
  }
}

myCor(nhcombined$Nettime_2018, nhcombined$Nettime_2017)

```



```
permCor(nhcombined$Nettime_2018, nhcombined$Nettime_2017)
```

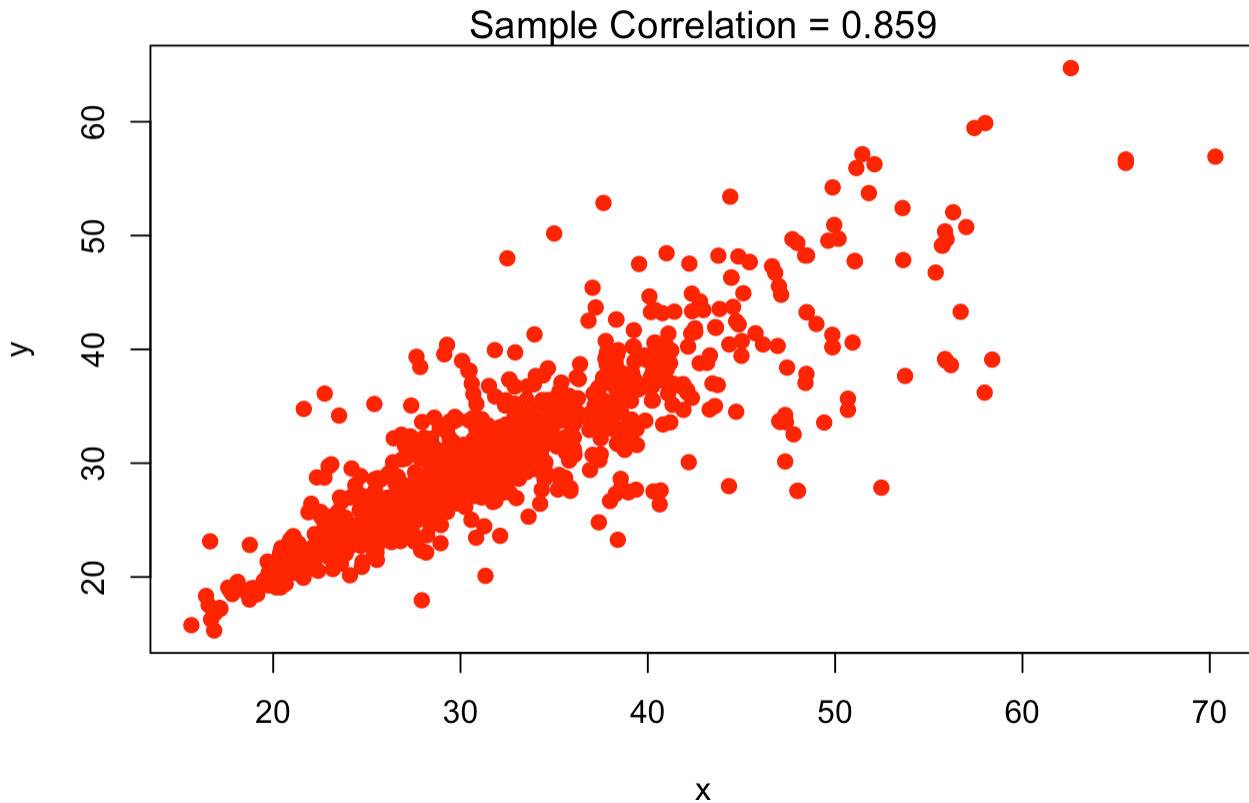


It seems appropriate to calculate correlation, even considering outliers. Looking at the myCor() scatterplot, there definitely

seems to be a linear trend, and this is reflected by the sample correlation = 0.879. Even if I looked at the scatterplot without the outlier in the top right, I would still be able to eyeball a similar trend. This result is further supported by the very low (essentially 0) p-value.

(1.3) Repeat part 1.2), but first remove the one individual who had times over 120 minutes in both years. Does it seem appropriate to calculate correlation now? How does the correlation change by removing the one outlying value? Is the correlation statistically significantly different from zero?

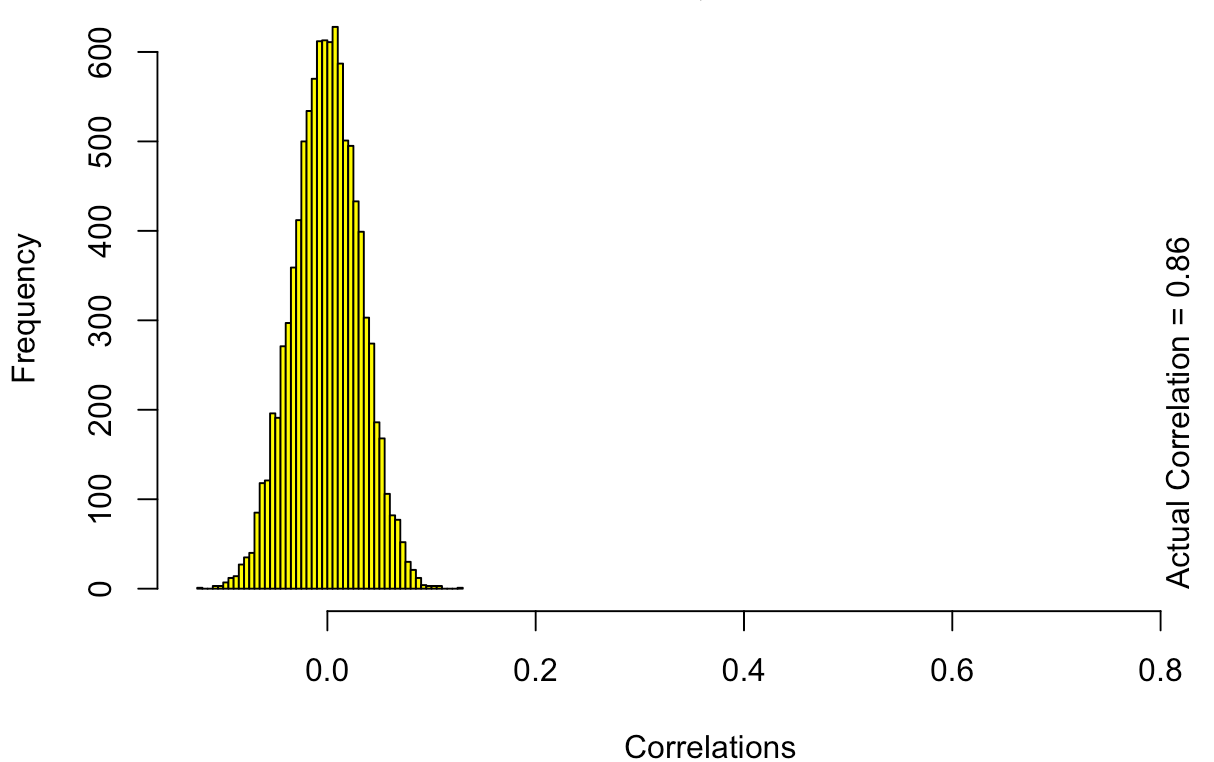
```
nh_no_outlier <- subset(nhcombined, !(Nettime_2018 > 120 & Nettime_2017 > 120))  
myCor(nh_no_outlier$Nettime_2018, nh_no_outlier$Nettime_2017)
```



```
permCor(nh_no_outlier$Nettime_2018, nh_no_outlier$Nettime_2017)
```

Permuted Sample Correlations

Permuted P-value = 0, Calculated P-value = 0



It still seems appropriate to calculate correlation, as the trend looks like it still exists, though to a lesser degree. This is supported by the sample correlation = 0.859, which is slightly less than what it was when the outlier was included (0.879). The correlation is statistically significantly different from zero because our p-value is still very low (essentially 0).

2) Correlation and the Bootstrap.

World Bank Data: CO2 Emissions per capita and Energy Use per capita

A .CSV file containing world bank data from 2016 can be downloaded [HERE](https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv) (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv).

The variables we'll consider in this example are:

- Country : countryname
- EnergyUse : kilograms of oil equivalent usage per capita
- CO2 : CO2 emissions per capita (metric tons)

In this problem, we will take a closer look at the correlation between Energy Use per capita and CO2 Emissions per capita.

(2.1) Read the data into an object called `wb`. Then make a dataframe called `wb2` that has complete data but only contains the country name, EnergyUse and CO2. Finally, create two new variables `logEnergy` and `logCO2` which are the natural log of each of these variables, respectively. You only need to retain these 5 columns in `wb2`.

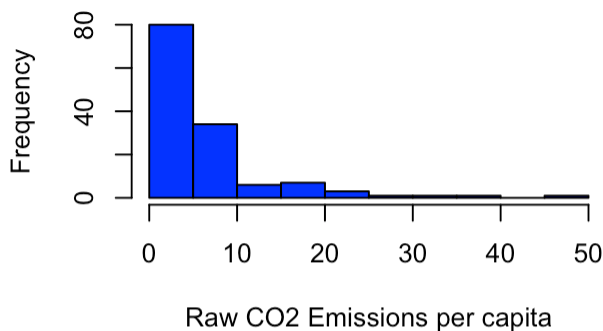
```
wb <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/WB.2016.csv")
wb2 <- subset(wb, select = c(Country, EnergyUse, CO2))
wb2 <- na.omit(wb2)
wb2$logEnergy <- log(wb2$EnergyUse)
wb2$logCO2 <- log(wb2$CO2)
head(wb2)
```

##	Country	EnergyUse	CO2	logEnergy	logCO2
## 2	Albania	808.4558	1.978763	6.695126	0.6824721
## 3	Algeria	1321.0995	3.717410	7.186220	1.3130272
## 6	Angola	545.0405	1.291328	6.300860	0.2556714
## 8	Argentina	2015.1870	4.746797	7.608467	1.5574702
## 9	Armenia	1018.0712	1.902759	6.925665	0.6433049
## 11	Australia	5328.2240	15.370138	8.580773	2.7324265

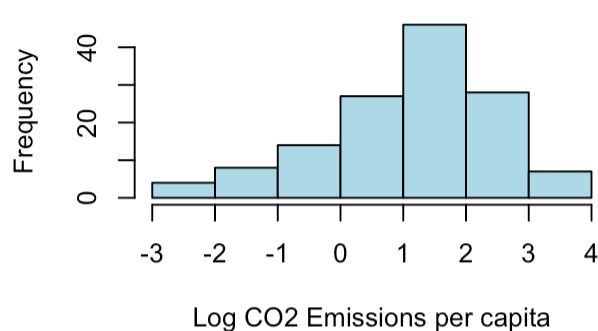
(2.2) Make histograms of CO2 and EnergyUse on both the raw scale and the log scale. Then make a scatterplot of CO2 (vertical axis) and EnergyUse (horizontal axis) on both the raw scale and the log scale. Put appropriate labels on all plots and make them look interesting (i.e. use a non-default color at least). Add the correlation to two decimal places between variables as a subtitle. Describe in a few words the shape of each distribution. Comment on the appropriateness of calculating correlation on both the raw scale and the log scale and comment on the correlation itself.

```
par(mfrow=c(2,2))
hist(wb2$CO2, xlab = "Raw CO2 Emissions per capita",
     main = "World Bank raw CO2 Emissions per capita", col = "blue")
hist(wb2$logCO2, xlab = "Log CO2 Emissions per capita",
     main = "World Bank log CO2 Emissions per capita", col = "lightblue")
hist(wb2$EnergyUse, xlab = "Raw Energy Use per capita",
     main = "World Bank raw Energy Use per capita", col = "green")
hist(wb2$logEnergy, xlab = "Log Energy Use per capita",
     main = "World Bank log Energy Use per capita", col = "lightgreen")
```

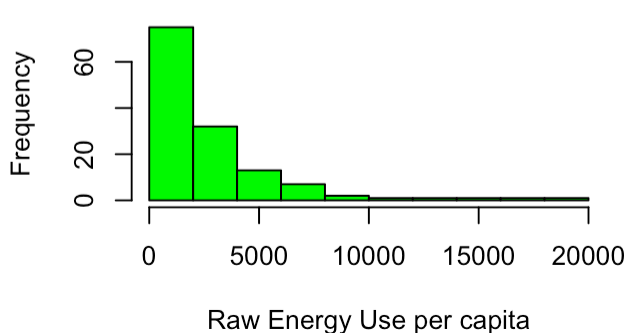
World Bank raw CO2 Emissions per capit



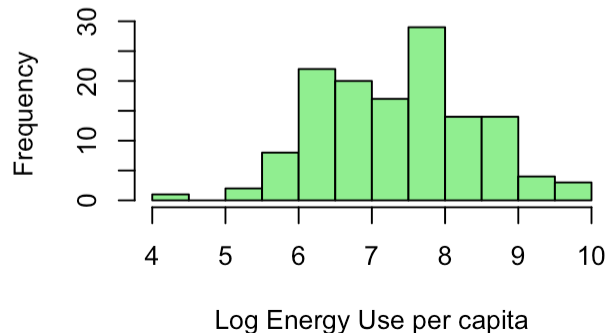
World Bank log CO2 Emissions per capita:



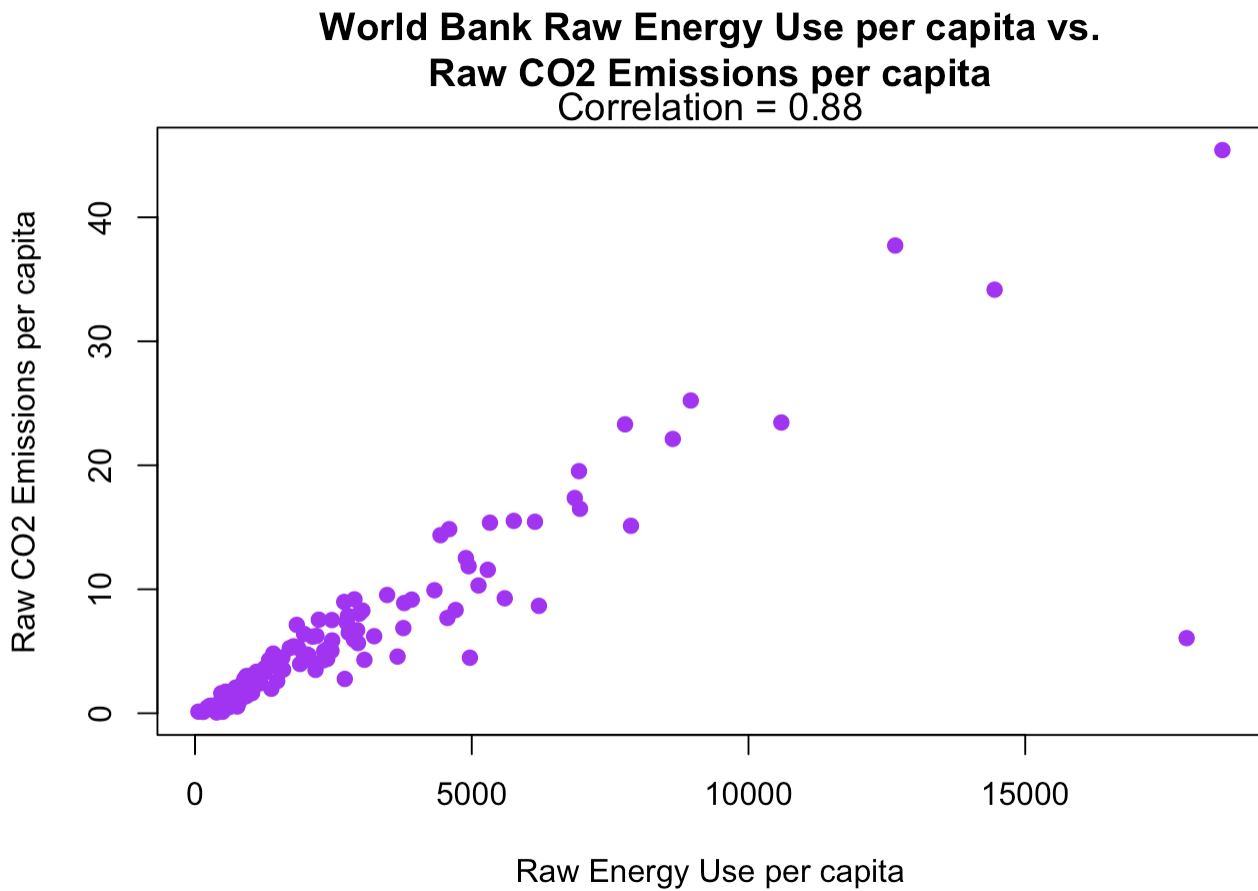
World Bank raw Energy Use per capita



World Bank log Energy Use per capita

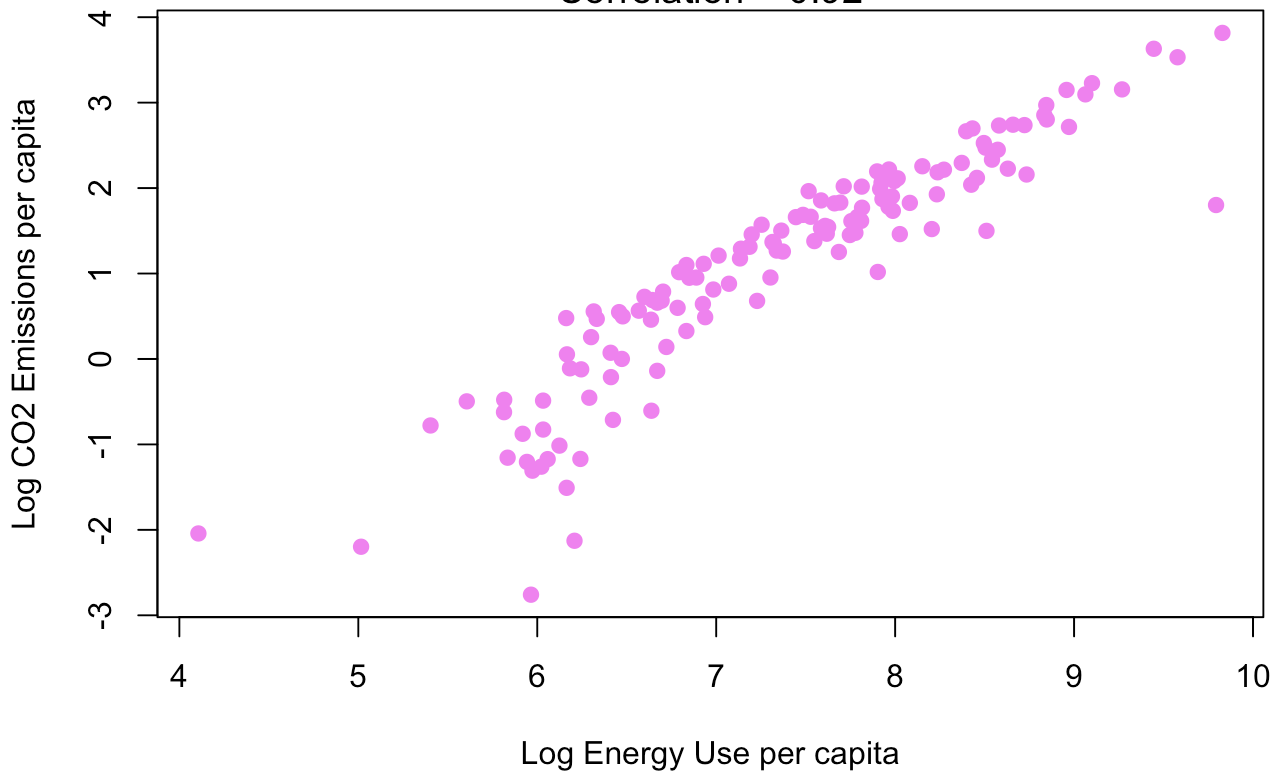


```
par(mfrow=c(1,1))
plot(wb2$EnergyUse, wb2$C02, xlab = "Raw Energy Use per capita", ylab = "Raw CO2 Emissions per capita",
      main = "World Bank Raw Energy Use per capita vs.\nRaw CO2 Emissions per capita", pch = 19, col = "purple")
mtext(paste("Correlation =", round(cor(wb2$EnergyUse, wb2$C02), 2)), side = 3, cex = 1.2)
```



```
plot(wb2$logEnergy, wb2$logC02, xlab = "Log Energy Use per capita", ylab = "Log CO2 Emissions per capita",
      main = "World Bank Log Energy Use per capita vs.\nLog CO2 Emissions per capita", pch = 19, col = "violet")
mtext(paste("Correlation =", round(cor(wb2$logEnergy, wb2$logC02), 2)), side = 3, cex = 1.2)
```

World Bank Log Energy Use per capita vs. Log CO2 Emissions per capita Correlation = 0.92



For both raw CO2 and raw energy, the distributions are heavily right skewed as most of the values are on the lower end. When you take the log of these values, the distributions tend to normalize a lot. As for the correlations, both raw and log correlations are quite high (0.88 and 0.92 respectively), and I assume this difference is most likely due to the outlier in the raw data that has very high energy use with very low CO2 emissions. I like the log correlation because the data points are more spread out compared to the raw correlation where they're concentrated in the lower end of the distribution. Generally, these correlations make sense to me since energy produced/used is associated with the burning of fossil fuels, resulting in more CO2 emissions.

(2.3) Identify the one country that is an outlier from the general trend (i.e. country with very HIGH Energy Usage and very LOW Emissions per capita). Give a one sentence reason why this country has the values it does (i.e. why is it an outlier from the trend).

```
wb2[wb2$EnergyUse > 15000 & wb2$C02 < 10, c("Country", "EnergyUse", "C02")]
```

```
##      Country EnergyUse      C02
## 88 Iceland  17916.12 6.059657
```

The country with very high energy usage and very low emissions per capita is Iceland. Iceland has these values because it largely relies on renewable energy, allowing for high energy use without producing a lot of CO2.

(2.4) Calculate (and display) a parametric test of the significance of the correlation between logCO2 and logEnergy, and save the result in an object called `cor1test`. Is the correlation statistically significantly different from zero? What is a 96% confidence interval for the location of the true correlation?

```
cor1test <- cor.test(wb2$logC02, wb2$logEnergy, conf.level = 0.96)
cor1test
```



```
##
## Pearson's product-moment correlation
##
## data: wb2$logCO2 and wb2$logEnergy
## t = 27.739, df = 132, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 96 percent confidence interval:
## 0.8927879 0.9462201
## sample estimates:
## cor
## 0.923886
```

logCO2 and logEnergy have a high correlation (0.923886) with a low p-value (< 2.2e-16), meaning that the correlation is statistically significantly different from zero. The 96% CI for the location of the true correlation is 0.8927879 to 0.9462201.

(2.5) Get 10000 bootstrap samples from `wb2` following the example given in Class 11. For each sample, calculate the correlation of `log(CO2)` and `log(Energy Use)` in your sample - save results in an object called `corResults`. Calculate a 96% bootstrap confidence interval for the true correlation based on quantiles. Save results in an objects called `ci_r` (for correlation). Display the results for these intervals.

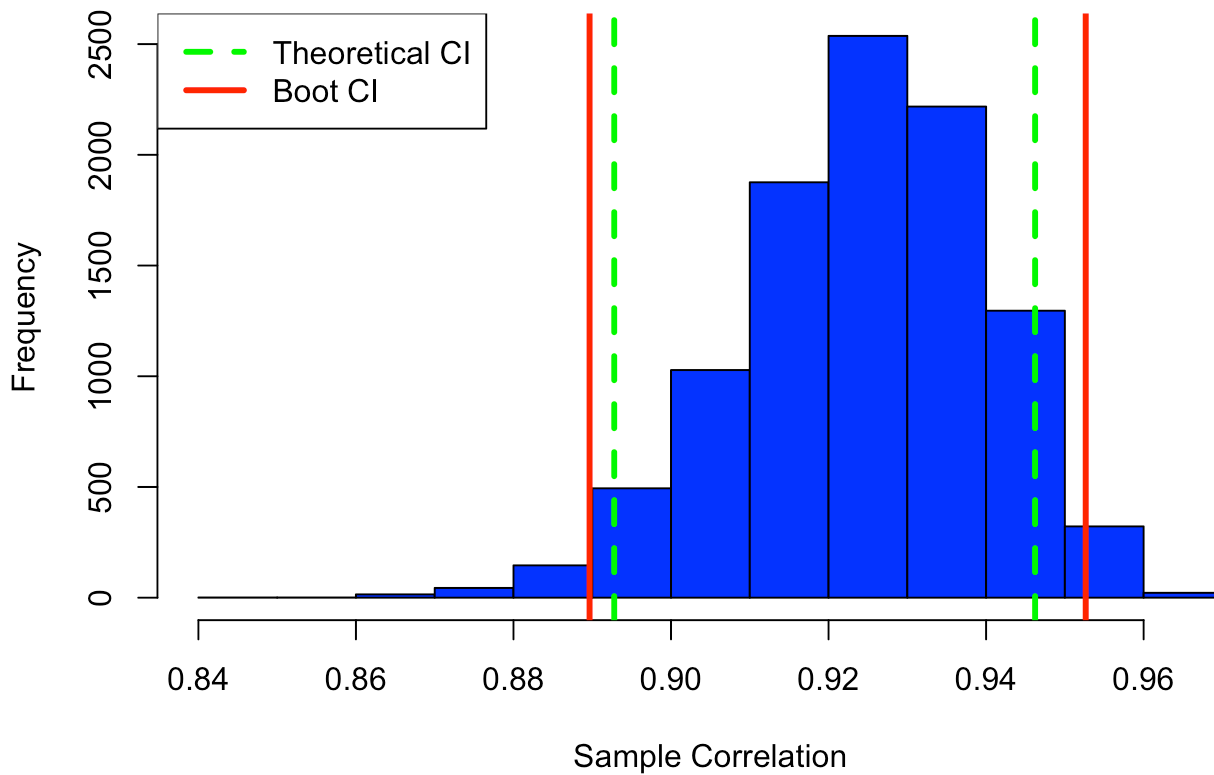
```
N <- nrow(wb2)
n_samp <- 10000
corResults <- rep(NA, n_samp)
for (i in 1:n_samp){
  s <- sample(1:N, N, replace = T)
  fakeData <- wb2[s, c("logCO2", "logEnergy")]
  corResults[i] <- cor(fakeData[, 1], fakeData[, 2])
}
ci_r <- quantile(corResults, c(0.02, 0.98))
ci_r
```

```
##          2%          98%
## 0.8896612 0.9526477
```

(2.6) Make a histogram of the results in `corResults`. Add vertical lines for both the theoretical confidence interval and the bootstrapped confidence interval.

```
hist(corResults, col = "blue", main = "Bootstrapped Correlations", xlab = "Sample Correlation")
abline(v = ci_r, lwd = 3, col = "red")
abline(v = cor1test$conf.int, lwd = 3, col = "green", lty = 2)
legend("topleft", c("Theoretical CI", "Boot CI"), lwd = 3, col = c("green", "red"), lty = c(2, 1))
```

Bootstrapped Correlations



(2.7) Comment on any differences you observe between the bootstrapped CI and the theoretical CI.

While both the bootstrapped CI and theoretical CI seem to be centered around the same point, the theoretical CI is slightly more narrow than the bootstrapped CI.

(2.8) Repeat parts 2.5 through 2.7 but on the RAW SCALE (i.e. use EnergyUse and CO2, not logEnergyUse and logCO2).

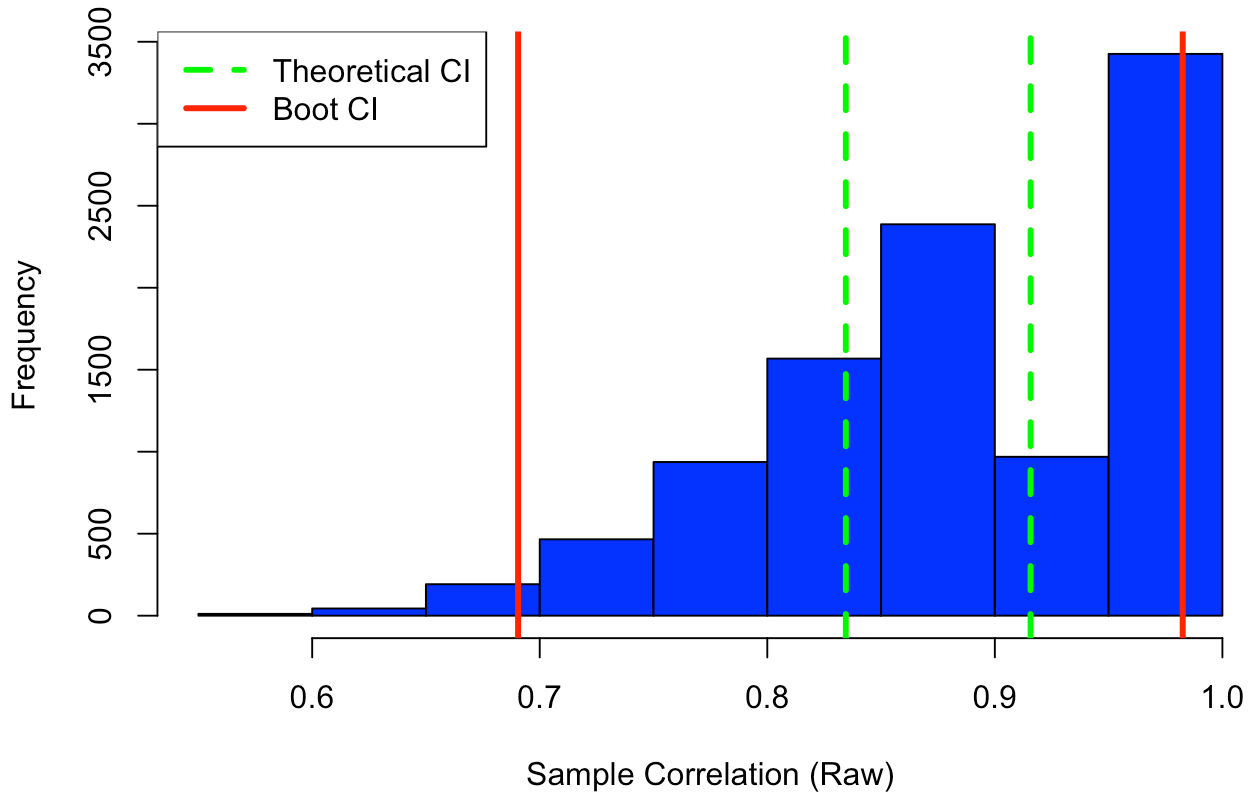
```
N <- nrow(wb2)
n_samp <- 10000
corResultsRaw <- rep(NA, n_samp)
for (i in 1:n_samp){
  s <- sample(1:N, N, replace = T)
  fakeData <- wb2[s, c("CO2", "EnergyUse")]
  corResultsRaw[i] <- cor(fakeData[, 1], fakeData[, 2])
}
ci_r_raw <- quantile(corResultsRaw, c(0.02, 0.98))
ci_r_raw
```

```
##          2%          98%
## 0.6904982 0.9825615
```

```
cor_raw_test <- cor.test(wb2$CO2, wb2$EnergyUse, conf.level = 0.96)

hist(corResultsRaw, col = "blue", main = "Bootstrapped Correlations (Raw)", xlab = "Sample Correlation (Raw)")
abline(v = ci_r_raw, lwd = 3, col = "red")
abline(v = cor_raw_test$conf.int, lwd = 3, col = "green", lty = 2)
legend("topleft", c("Theoretical CI", "Boot CI"), lwd = 3, col = c("green", "red"), lty = c(2, 1))
```

Bootstrapped Correlations (Raw)



In the raw scale, we see that both the CIs are still centered around a similar area (around the first peak), but the bootstrapped CI is MUCH larger than the theoretical CI.

(2.9) Repeat 2.8, but without Iceland. Write a few sentences on what changes.

```
wb2_no_Iceland <- subset(wb2, Country != "Iceland")

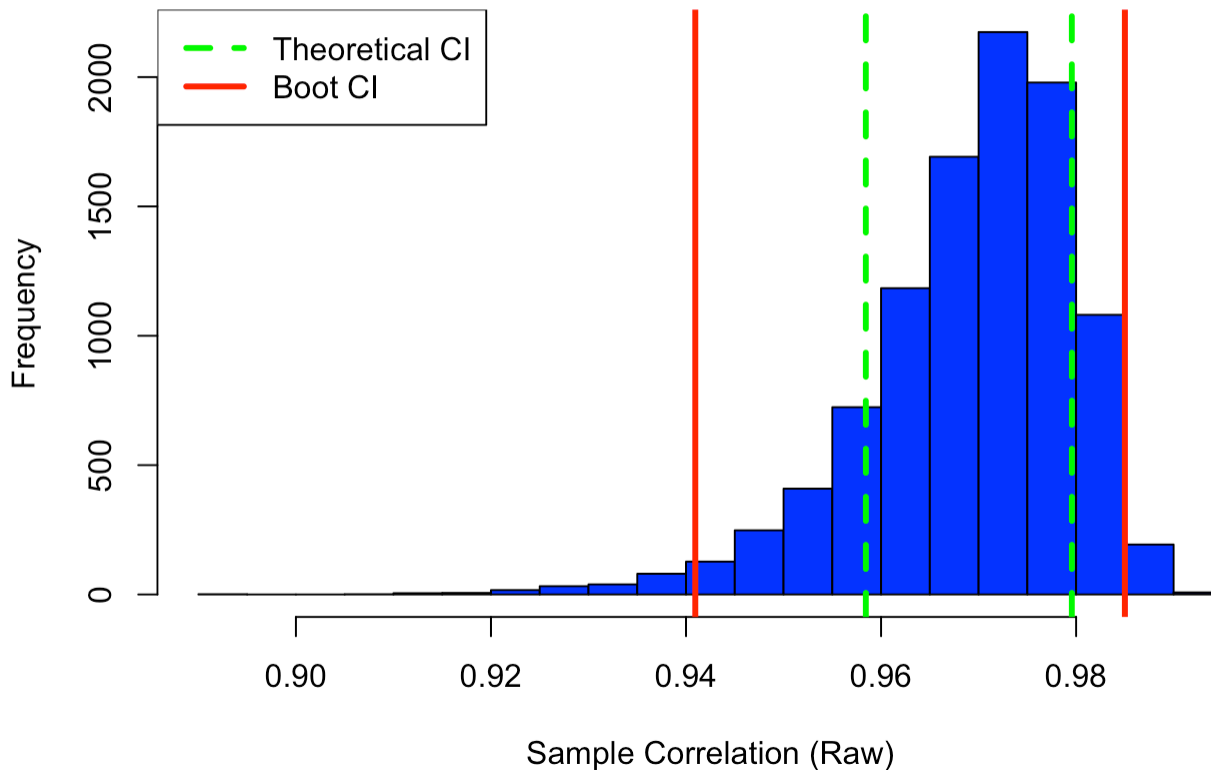
N <- nrow(wb2_no_Iceland)
n_samp <- 10000
corResultsRawNoIceland <- rep(NA, n_samp)
for (i in 1:n_samp){
  s <- sample(1:N, N, replace = T)
  fakeData <- wb2_no_Iceland[s, c("CO2", "EnergyUse")]
  corResultsRawNoIceland[i] <- cor(fakeData[, 1], fakeData[, 2])
}
ci_r_raw_no_Iceland <- quantile(corResultsRawNoIceland, c(0.02, 0.98))
ci_r_raw_no_Iceland
```

```
##          2%          98%
## 0.9409511 0.9850015
```

```
cor_raw_no_Iceland_test <- cor.test(wb2_no_Iceland$CO2, wb2_no_Iceland$EnergyUse, conf.level = 0.96)
```

```
hist(corResultsRawNoIceland, col = "blue", main = "Bootstrapped Correlations (Raw, no Iceland)",
     xlab = "Sample Correlation (Raw)")
abline(v = ci_r_raw_no_Iceland, lwd = 3, col = "red")
abline(v = cor_raw_no_Iceland_test$conf.int, lwd = 3, col = "green", lty = 2)
legend("topleft", c("Theoretical CI", "Boot CI"), lwd = 3, col = c("green", "red"), lty = c(2, 1))
```

Bootstrapped Correlations (Raw, no Iceland)



Here, we can see that the distribution changed significantly from 2.8. While still left skewed, the distribution without Iceland only has one peak now. For the CIs, we see that without Iceland, the CIs are generally much tighter than before. And while there is still a visible difference between the theoretical and bootstrapped CIs when excluding Iceland, it is much less of a difference than when Iceland was included.

3) Class Survey Data

This questions involves cleaning and making bivariate plots of several variables from the class survey data. I'm going to be less specific of what you need to do - I'll say that at this point you have all the tools you need to clean the data and make interesting plots.

Just run the code below to get the data and remind you of the variables that exist.

```
#Get data
survey <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/class.survey.csv", header = TRUE, as.is = T)
names(survey)
```

```
## [1] "ClassProb" "Status" "Year" "Division" "Gender"
## [6] "HtCm" "Hand" "Haircut" "Exercise" "Coursework"
## [11] "Web" "TV" "Social" "Econ" "Animal"
## [16] "Friends" "Pulse"
```

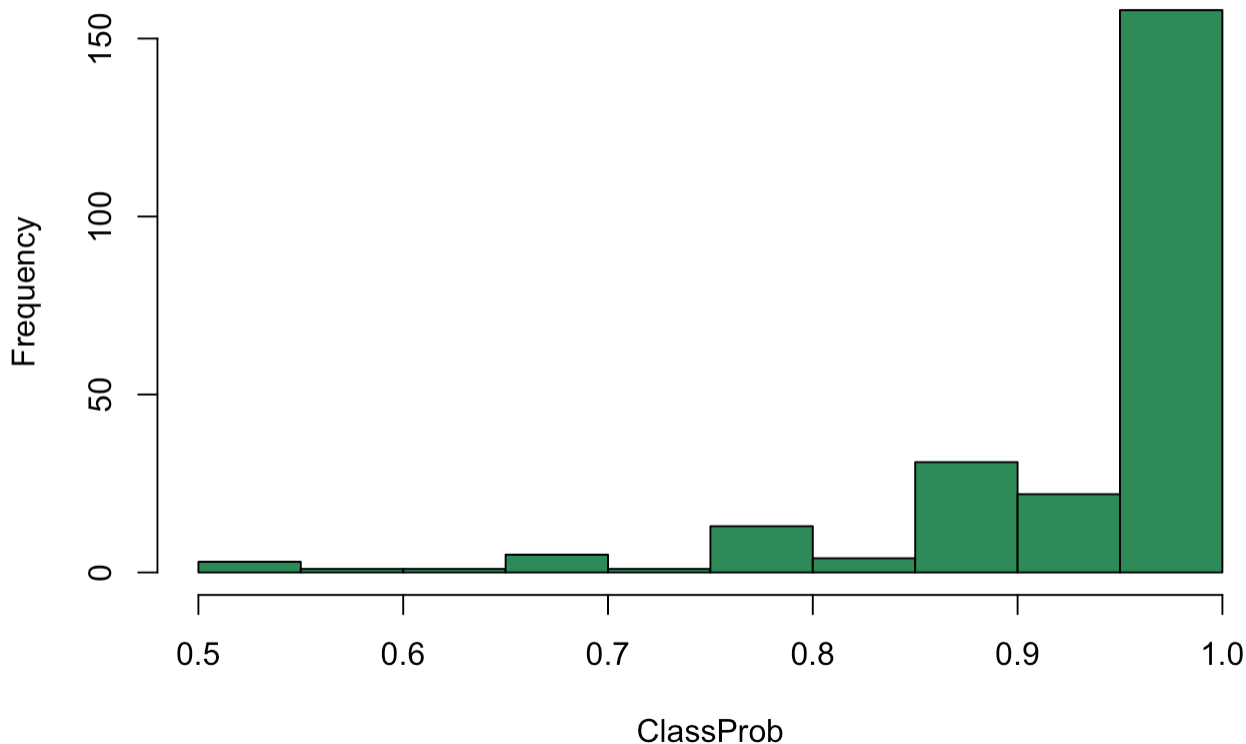
```
dim(survey)
```

```
## [1] 239 17
```

(3.1) Clean the variable `ClassProb` so that all values are between zero and 1. Assume that any value over 20 is in a percentage and should be changed to a probability. Any value more than 1 but less than 20 should be changed to `NA`. Make a nice looking histogram of the resulting data and comment on the shape of the distribution.

```
survey$ClassProbClean <- survey$ClassProb
survey$ClassProbClean[survey$ClassProbClean > 20] <- survey$ClassProbClean[survey$ClassProbClean
>= 20] / 100
survey$ClassProbClean[survey$ClassProbClean > 1 & survey$ClassProbClean < 20 ] <- NA
hist(survey$ClassProbClean, xlab = "ClassProb", main = "Histogram of Cleaned Class Probability",
col = "seagreen")
```

Histogram of Cleaned Class Probability



The shape of the histogram seems heavily left skewed with most of the values being above 0.85.

(3.2) Clean the variables `Social` and `Econ` so that they are integer and numeric on the scale 1 to 7. Make a two-way table of the results. Does it appear there is a relationship between these variables?

```
# Just to get an idea of the values we're working with
str(survey$Social)
```

```
## chr [1:239] "4 = Moderate" "5" "5" "6" "6" "6" "6" "6" "6" "4 = Moderate" ...
```

```
unique(survey$Social)
```

```
## [1] "4 = Moderate"      "5"  
## [3] "6"                  "7 = Extremely Liberal"  
## [5] "3"                  "No answer text provided."  
## [7] "2"                  "1 = Extremely Conservative"
```

```
str(survey$Econ)
```

```
## chr [1:239] "4 = Moderate" "5" "5" "5" "4 = Moderate" "5" "5" "3" "5" ...
```

```
unique(survey$Econ)
```

```
## [1] "4 = Moderate"      "5"  
## [3] "3"                  "6"  
## [5] "2"                  "7 = Extremely Liberal"  
## [7] ""                   "1 = Extremely Conservative"
```

```
survey$SocialClean <- as.integer(sub(".*", "", survey$Social))
```

```
## Warning: NAs introduced by coercion
```

```
survey$EconClean <- as.numeric(sub(".*", "", survey$Econ))
```

```
# Check to see if it was properly cleaned  
str(survey$SocialClean)
```

```
## int [1:239] 4 5 5 6 6 6 6 6 6 4 ...
```

```
unique(survey$SocialClean)
```

```
## [1] 4 5 6 7 3 NA 2 1
```

```
str(survey$EconClean)
```

```
## num [1:239] 4 5 5 5 4 5 5 3 5 4 ...
```

```
unique(survey$EconClean)
```

```
## [1] 4 5 3 6 2 7 NA 1
```

```
table(survey$SocialClean, survey$EconClean)
```

```
##
##      1  2  3  4  5  6  7
##    1  2  0  0  0  0  0
##    2  0  0  0  1  2  1
##    3  0  4  5  3  2  0
##    4  0  0  9 29  5  1
##    5  0  1 13 20 21 10
##    6  0  1  2 21 29 22
##    7  0  0  0  0  7  5
```

```
cor.test(survey$SocialClean, survey$EconClean, use = "complete.obs")
```

```
##
## Pearson's product-moment correlation
##
## data: survey$SocialClean and survey$EconClean
## t = 11.053, df = 232, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.4964606 0.6654378
## sample estimates:
##      cor
## 0.5873129
```

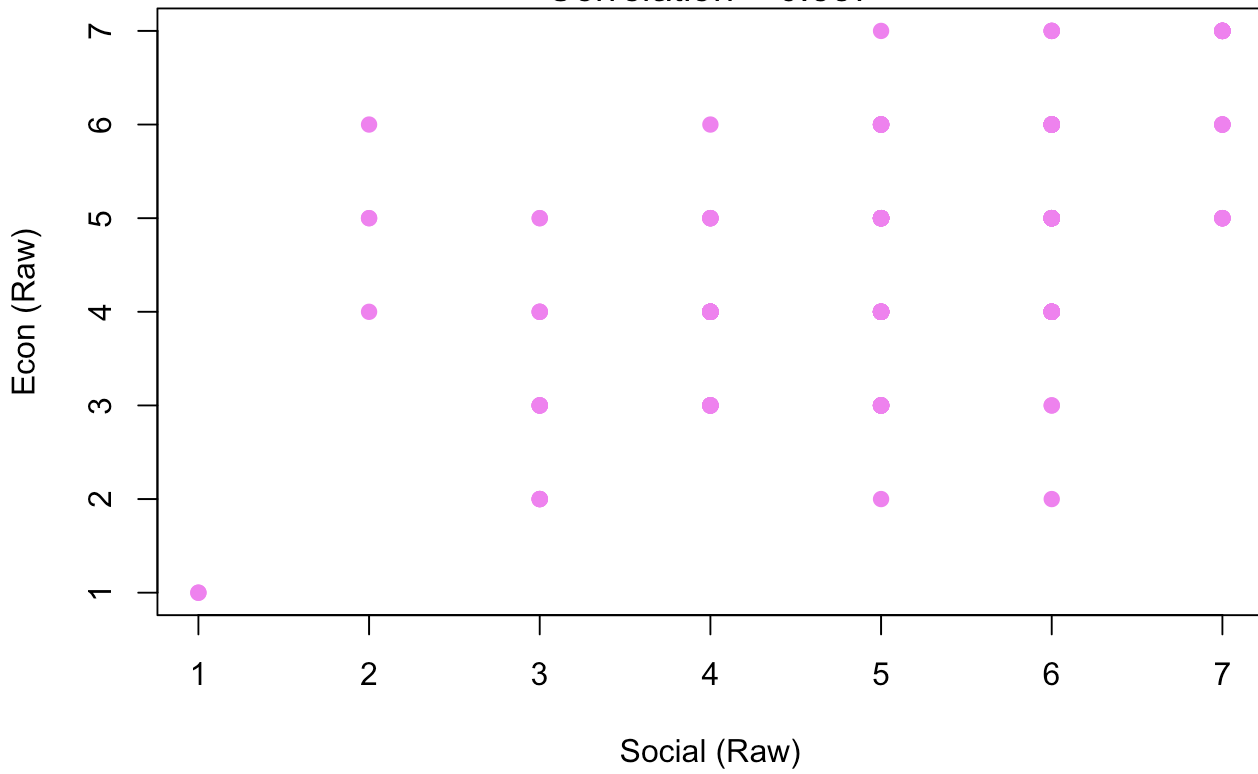
Based on the table alone, we can see that there is a generally positive trend between Social and Econ. Doing a correlation test, we can see that there is a moderately positive relationship between Social and Econ ($cor = 0.587$), and it is significant because our p -value is $< 2.2e-16$.

(3.3) Make three scatterplots of Social and Econ - one on the raw scale, one with jittered results, one with radii proportional to the square root of the frequency. Follow the examples shown in Class 11. Each plot should include the correlation in the labels.

```
# Raw scale
social <- survey$SocialClean
econ <- survey$EconClean
plot(social, econ, xlab = "Social (Raw)", ylab = "Econ (Raw)",
     main = "Social (Raw) vs. Econ (Raw)", pch = 19, col = "violet")
mtext(paste("Correlation =", round(cor(social, econ, use = "complete.obs"), 3)), side = 3, cex =
1.2)
```

Social (Raw) vs. Econ (Raw)

Correlation = 0.587



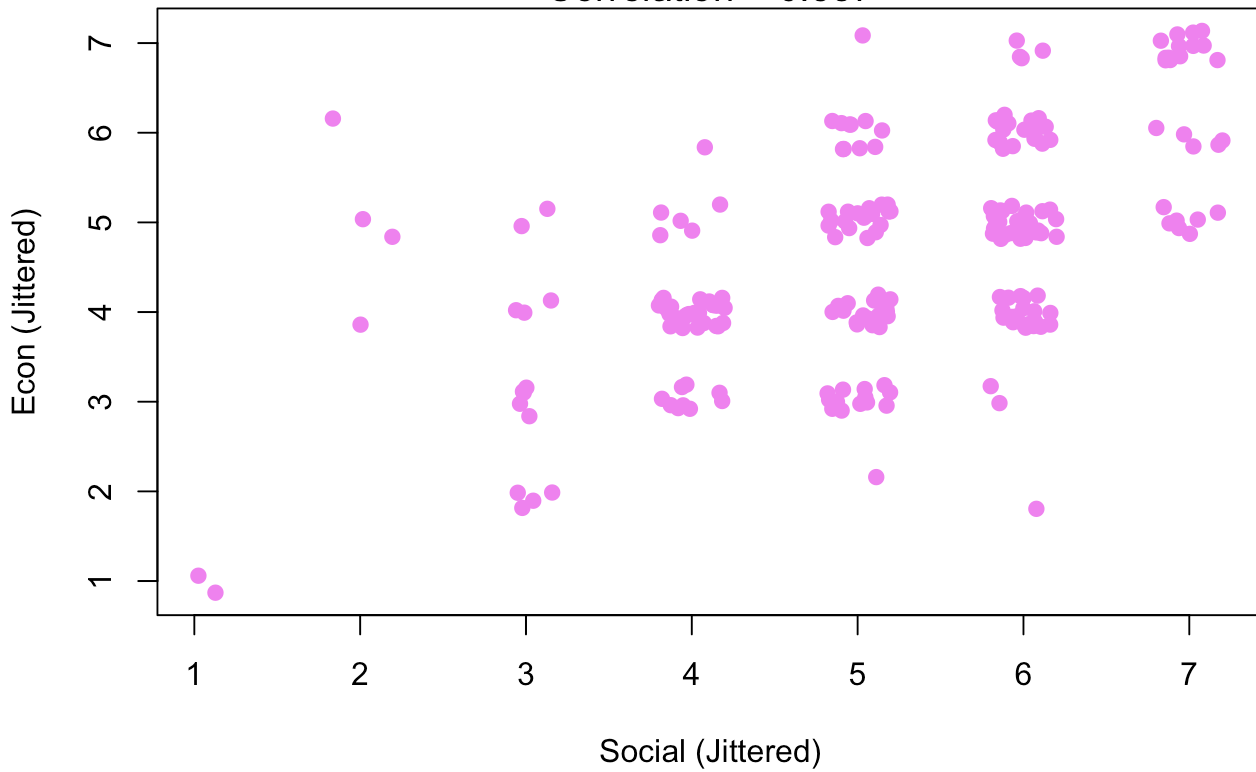
```
# Jittered results
```

```
plot(jitter(social), jitter(econ), xlab = "Social (Jittered)", ylab = "Econ (Jittered)",  
     main = "Social (Jittered) vs. Econ (Jittered)",  
     pch = 19, col = "violet")
```

```
mtext(paste("Correlation =", round(cor(social, econ, use = "complete.obs"), 3)), side = 3, cex =  
1.2)
```

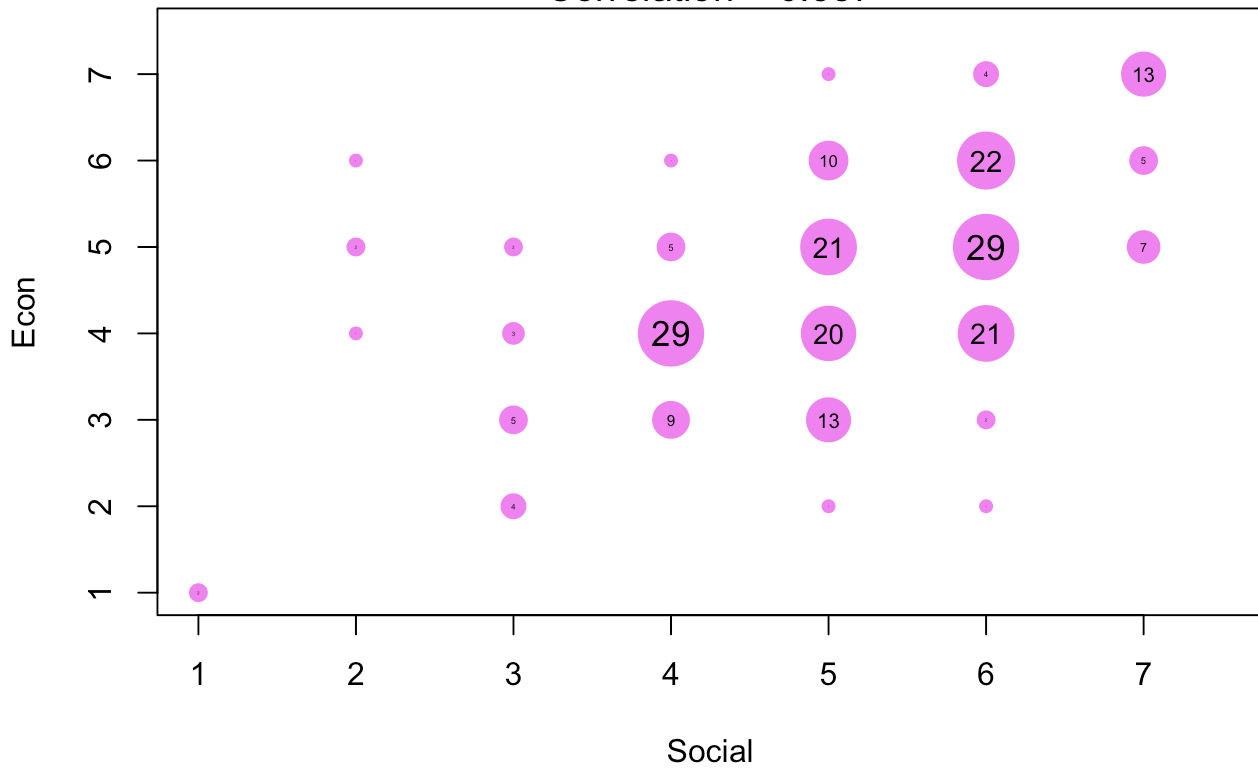

Social (Jittered) vs. Econ (Jittered)

Correlation = 0.587



```
# Radii proportional to the square root of the frequency
freq <- (c(table(survey$SocialClean, survey$EconClean)))
x1 <- rep(c(1:7), 7)
y1 <- sort(x1)
plot(x1, y1,
     pch = 19,
     col = "violet",
     xlab = "Social",
     ylab = "Econ",
     main = "Social vs. Econ\n(Radii proportional to the square root of the frequency)",
     cex = sqrt(freq/max(freq)*20),
     ylim = c(1, 7.5),
     xlim = c(1, 7.5))
mtext(paste("Correlation =", round(cor(social, econ, use = "complete.obs"), 3)), side = 3, cex =
1.2)
freq_labels <- ifelse(freq == 0, NA, freq)
text(x1, y1, freq_labels, cex = log(freq/max(freq)*2 + 1), col = "black")
```

Social vs. Econ
(Radii proportional to the square root of the frequency)
 Correlation = 0.587



(3.4) Write a quick function called `cleanIt()` that cleans the variables `Exercise` and `Coursework` so that they are numeric. Apply this function to these two variables (which by the way are in minutes) and make histograms of the results. In a sentence, describe what you see.

```
str(survey$Exercise)
```

```
##  num [1:239] 700 300 65 80 30 120 90 90 60 40 ...
```

```
unique(survey$Exercise)
```

```
##  [1] 700.0 300.0 65.0 80.0 30.0 120.0 90.0 60.0 40.0 180.0 360.0 100.0
## [13] 220.0 18.0 240.0  NA 200.0 420.0 500.0  0.0  1.0 600.0 331.0 455.0
## [25] 630.0 225.0 250.0 50.0 445.0 210.0  2.5 650.0 900.0 720.0 150.0 400.0
## [37] 45.0  7.0 840.0 450.0 20.0 390.0 800.0 480.0 540.0 70.0 440.0 160.0
## [49] 930.0 140.0 15.0  2.0 750.0  5.0 75.0
```

```
str(survey$Coursework)
```

```
##  chr [1:239] "120" "300" "120" "180" "180" "180" "180" "180" "180" "150" "180" ...
```

```
unique(survey$Coursework)
```

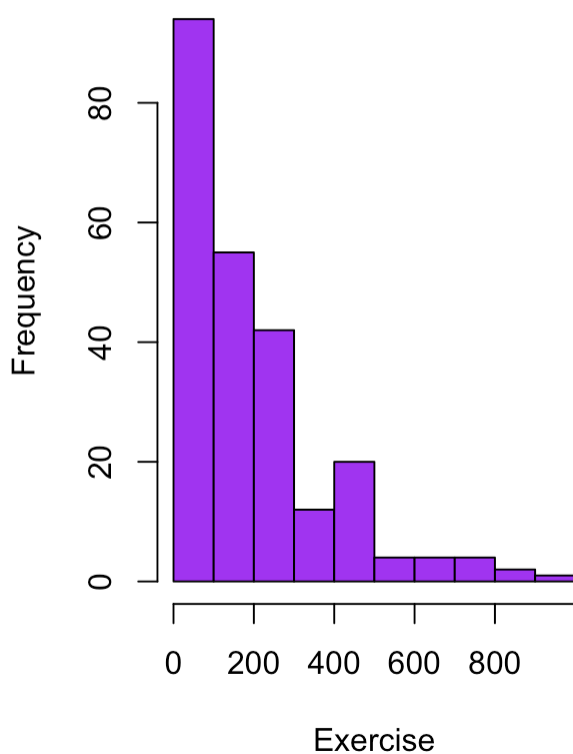
```
## [1] "120" "300" "180" "150" "360" "60" "200" "400" "3"
## [10] "330" "2,100" "220" "240" "250" "1,800" "100" "210" "140"
## [19] "367" "160" "30" "350" "90" "270" "6" "600" "45"
## [28] "480" "5" "75" "450" "70" "1,260" "8" "1,380"
```

```
cleanIt <- function(x){
  x <- as.numeric(gsub(",", "", trimws(x)))
  return (x)
}

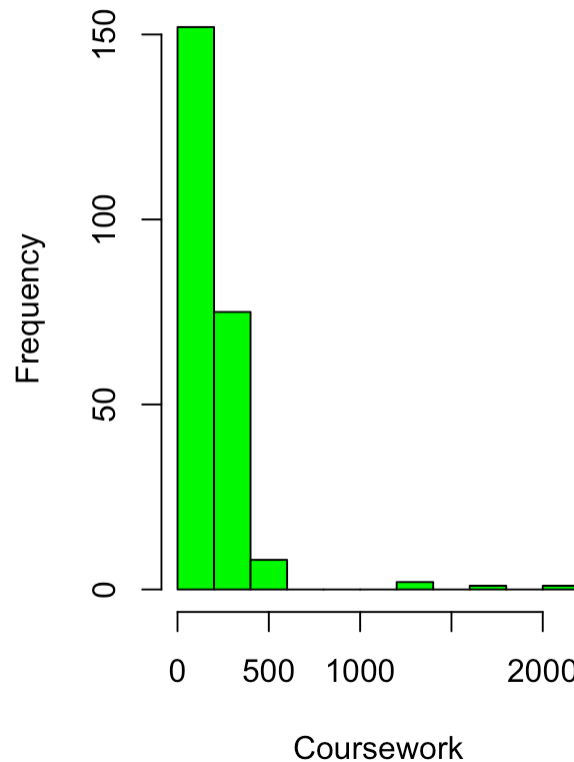
survey$ExerciseClean <- cleanIt(survey$Exercise)
survey$CourseworkClean <- cleanIt(survey$Coursework)

par(mfrow = c(1,2))
hist(survey$ExerciseClean, xlab = "Exercise", main = "Exercise histogram", col = "purple")
hist(survey$CourseworkClean, xlab = "Coursework", main = "Coursework histogram", col = "green")
```

Exercise histogram



Coursework histogram



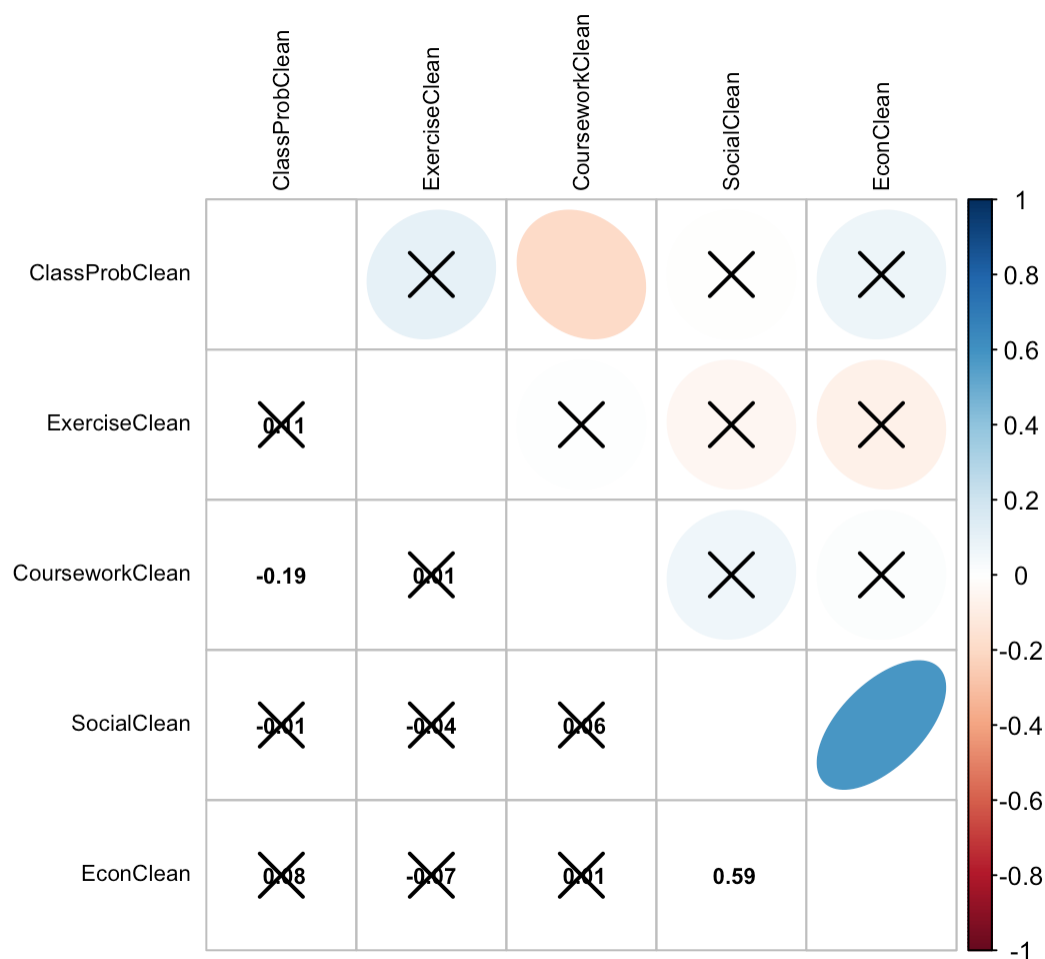
In the histograms, the shapes between Exercise and Coursework are pretty similar (both heavily right skewed). However, it seems that exercise is more broadly spread out, as it has lower peak frequency and more columns that do not have a frequency = 0.

(3.5) Following code in classes 11 and 12, create plots `corrplot.mixed` and `chart.Correlation()` for all the variables mentioned in parts 3.1 through 3.4. Write not more than two sentences about what you observe and what you might do next. Note: ignore any warnings you might get, OR just include the cod ``warning = F`` at the top of your chunk.

```
library(corrplot)
```

```
## corplot 0.95 loaded
```

```
cols <- c("ClassProbClean", "SocialClean", "EconClean", "ExerciseClean", "CourseworkClean")
sigcorr <- cor.mtest(survey[, cols], conf.level = 0.95)
corrplot.mixed(
  cor(survey[, cols], use = "complete.obs"), lower.col = "black", upper = "ellipse", tl.col = "black",
  number.cex = 0.7, order = "hclust", tl.pos = "lt", tl.cex = 0.7, p.mat = sigcorr$p, sig.level =
0.05)
```



```
library(PerformanceAnalytics)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

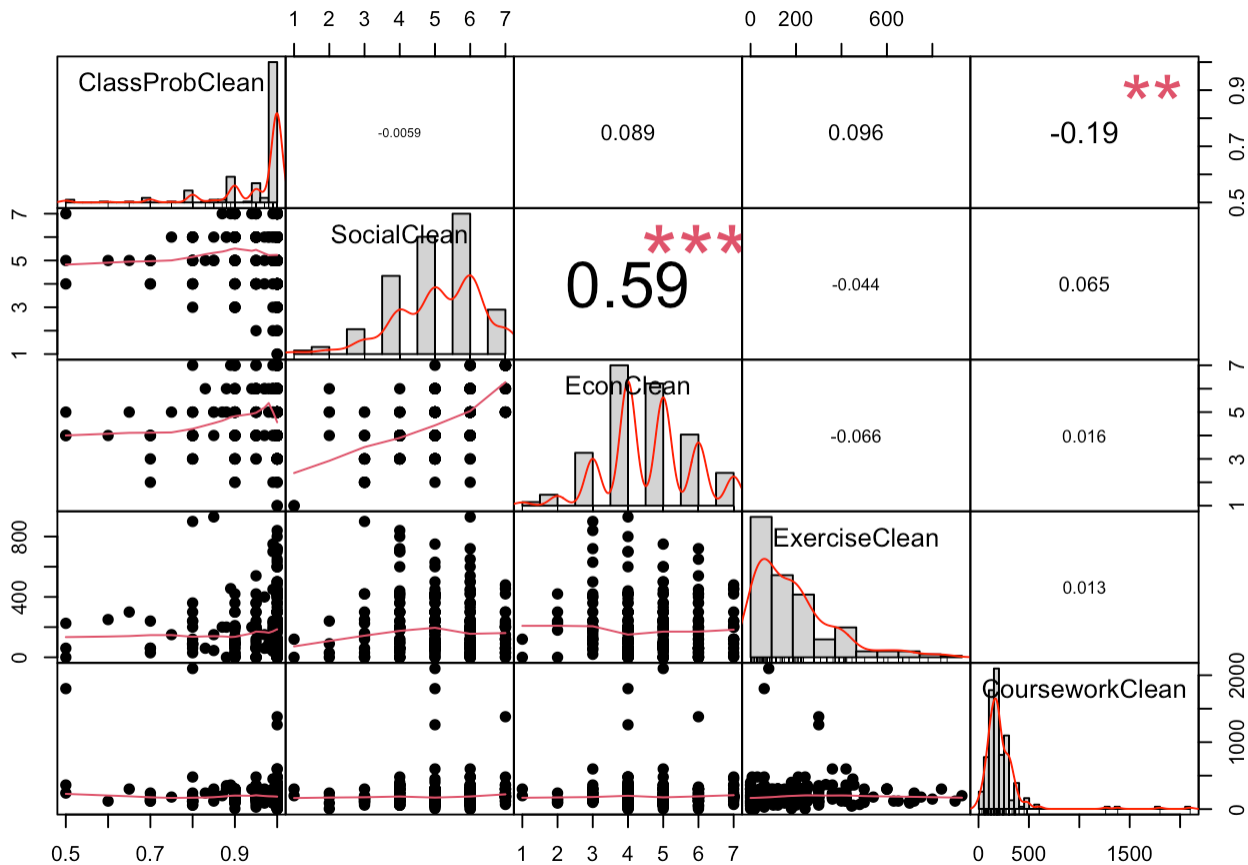
```
##  
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':  
##  
## as.Date, as.Date.numeric
```

```
##  
## Attaching package: 'PerformanceAnalytics'
```

```
## The following object is masked from 'package:graphics':  
##  
## legend
```

```
chart.Correlation(survey[, cols], histogram = TRUE, pch = 19)
```



Based on the output, I would say that the two correlations of interest are the positive correlation between Social and Econ and the negative correlation between Coursework and ClassProb. Next, I might try bootstrapping to find some CIs to see if a correlation of 0 is within the 95% CI.